

Health measures for the prevention project

Construction, validation, and the choice among them

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Built and validated by `data_cleaning/03--06` and `descriptives/01--06`; every number here is printed by those scripts' self-validation output. This note records which UKHLS health variables the project uses, how each measure is constructed and why, what the new graded-response measures deliver against everything that came before them (the original physical GRM, PCS/MCS, the physical-only composite, SF-6D, GHQ), and how the ever-diagnosis, dimensionality and ceiling questions resolve.

1 The variable inventory

Measurement models are restricted to items asked in (nearly) every wave, so a score means the same thing wherever it lands in the panel. Presence was verified against the raw headers of all fifteen `indresp` files, labels against the UKDA-6614 dictionaries.

Variables	Waves	Used
SF-12, all 12 items (self-completion preferred)	all 15	both GRMs; PCS/MCS re-build; SF-6D
GHQ-12 items (<code>scghqa..1</code> , 4 categories)	all 15	mental GRM
Impairment list <code>disdif1..12</code> , gated on <code>health</code>	all 15	physical GRM (FUNC testlet)
Long-standing illness (<code>health</code>)	all 15	gate for <code>disdif</code> ; descriptive
Diagnosed conditions (<code>hcond*</code> families, codes 1–17)	all 15 (§2)	condition items; sensitivities
Health satisfaction (<code>sclfsat1</code>)	all 15	validation outcome only
SWEMWBS; ADL module; loneliness	1/4/7/10/13; 7/9/11/13; 9–15	excluded: fail the every-wave rule
GP visits, hospital admissions	7–15	criterion-validity outcomes
Height/weight (self-report)	wave 1 only	excluded (biomarkers are UKDA-7251, not held)

2 Chronic conditions

2.1 Construction

UKHLS asks the full “ever diagnosed” inventory (`hcond1..17`) at a person's *first* interview; later waves record conditions newly reported since the last interview under a name that changes with

questionnaire redesigns — `hcondn{i}` (waves 2–9), `hcondcode{i}` (10–12), `hcondncode{i}` (13–15), all dictionary-verified. The running indicator (*ever had condition i by wave w $\Leftrightarrow$ reported in any family at any wave $\leq w$*) is monotone and defined from the first inventory onward (76% of person-waves). Only the wave-1 code list is used, so the definition is identical in every wave. Per-condition indicators, diagnosis ages and group counts live in `chronic_conditions.parquet`; nothing is collapsed before modelling.

This corrects two defects in the earlier EIT count, which used `hcondns*` (“still has *newly diagnosed* condition”) as its new-report family: that family has no codes 6 and 7 at all — every in-panel heart attack and stroke was missed — and it drops new diagnoses reported as already resolved. Our count is $\geq$ the EIT count on 100.000% of 458k person-waves (exact on 94.2%; 26,516 documented additions, concentrated in MI, stroke and cancer).

2.2 Why “ever”, not “current”

The “still has” family (`hconds##`) is asked *only at the wave a condition is reported*: at wave 6, of 5,090 ever-asthma respondents, $\sim$ 570 were routed to it. A per-wave “current” variable would be $\sim$ 90% carry-forward imputation. Beyond measurability, self-reported remission conflates control with cure — managed hypertension and diabetes read as “no longer have”, the mechanism behind the under-reporting found against measured disease (Johnston, Propper & Shields 2009) and GP records (Kriegsman et al. 1996: good agreement for diabetes, poor for heart disease and arthritis). The monotone ever-indicator matches the deficit-accumulation tradition (Mitnitski, Mogilner & Rockwood 2001; Searle et al. 2008). Its genuine cost — old diagnoses shadowing current health — is quantified in §5, not assumed away.

2.3 Why grouped, not seventeen indicators

Grouping is the testlet logic applied to diagnoses. CHD, angina, MI, CHF and stroke are one disease process reported several ways; as separate items they would violate local independence exactly as the `sf2a/sf2b` pair did before the original GRM collapsed them (Wainer & Kiely 1987; Yen 1993). Rare conditions (1–2% prevalence) also make weak, unstable IRT items — minimum prevalence is an explicit deficit-inclusion criterion in Searle et al. (2008) — and groups keep a stable clinical meaning while the code list grows (cf. Barnett et al. 2012). An ordinal within-group count (0/1/2+) keeps a severity gradient the way $PF = sf2a + sf2b$ does. Groups: **CVD** {CHF, CHD, angina, MI, stroke}, **METAB** {diabetes, hypertension}, **RESP** {asthma, bronchitis, emphysema}, **MSK** {arthritis}, **CANCER**, **OTHER** {thyroid, liver, epilepsy}; **DEPR** {clinical depression} sits on the mental side. Ever-prevalence: hypertension 21.4%, arthritis 16.2%, depression 8.2%, diabetes 7.2%, any CVD event 7.4%, cancer 5.1%.

2.4 Diagnosis timing

Every inventory report carries `hconda##` (age told); every in-panel report is dated by its wave: **100.0% of ever-flags carry a usable diagnosis age**, so recency-restricted variants are exact. The stock is old: median diagnosis age 11 years ($p_{25} = 5$, $p_{75} = 21$); at ages 65–80, 54% of flags are >10 years old, 26% >20 .

3 The graded-response measures

3.1 Why separate physical and mental banks

The measurement model must stay a static map from responses to a score (dynamics are estimated afterwards), and PROMIS practice is separate unidimensional banks per domain (Cella et al. 2010). Two clean banks also make the physical–mental correlation an *estimate* rather than an artifact: the SF-12’s orthogonal scoring forces $r(\text{PCS}, \text{MCS}) = 0$; our corrected promax on the subscales gives 0.59; published correlated-scoring estimates are 0.62 (Farivar, Cunningham & Hays 2007) and 0.71–0.73 (Tucker et al.); the two GRM thetas correlate **0.457**. Cross-loading SF-12 items are assigned by this repo’s own UK wave-1 varimax loadings (phys/ment): PF .86/.14, RP .85/.25, BP .75/.17, GH .72/.27 $\rightarrow$ physical; MH .09/.89, RE .26/.82, SF .48/.64 $\rightarrow$ mental. **VT (energy, .55/.44) enters neither bank** (PROMIS likewise treats fatigue as its own domain); cognitive and sensory disdif items are separate WHODAS-type constructs and are excluded from both.

3.2 The banks

Every item is coded so category 1 is worst health.

Spec	Items
P-FUNC	GH, PF, RP, BP (the validated original testlets) + FUNC: count of physical disdif limitations {mobility, lifting, dexterity, coordination, personal care}, 0/1/2/3+; health =“no” is a valid structural zero (module skipped by design)
P-FULL	P-FUNC + six ever-diagnosis group items (CVD/METAB/RESP 0/1/2+; MSK/CANCER/OTHER binary)
P-REC	as P-FULL, counting only diagnoses within the last 10 years
P-TIMED	as P-FULL with each group split into recent ($\leq 10\text{y}$) and stale ($> 10\text{y}$) items — the model <i>estimates</i> the recency weighting (§5)
MENT	MH ((6 – sf6a) + sf6c – 1, 9 levels), RE, SF, GHQ-positive and GHQ-negative wording testlets (19 levels each), ever-depression
COMBINED	P-FULL $\cup$ MENT: all seventeen good-coverage items in one bank (§6)

GHQ-12 is validated internationally (Goldberg et al. 1997) but its factor-analytic “dimensions” largely reflect negative-wording method effects (Hankins 2008, contra Graetz 1991). The pre-specified rule — fit item-level; collapse by wording if Yen’s Q_3 shows positive within-wording dependence — *triggered* (within-wording Q_3 max +0.26), the same cure the original GRM applied to the PF/RP pairs.

3.3 Estimation and diagnostics

Same machinery as the exactly-replicated original (Samejima 1969; marginal ML by EM over response patterns, 121-point quadrature, pooled latent $N(0, 1)$; multigroup by single year of age 20–90 for μ_a, σ_a). Samples: P-FUNC 497,423 person-years; P-FULL/P-REC/P-TIMED 387,599 (condition items exist from first inventory); MENT 379,462; COMBINED 376,019.

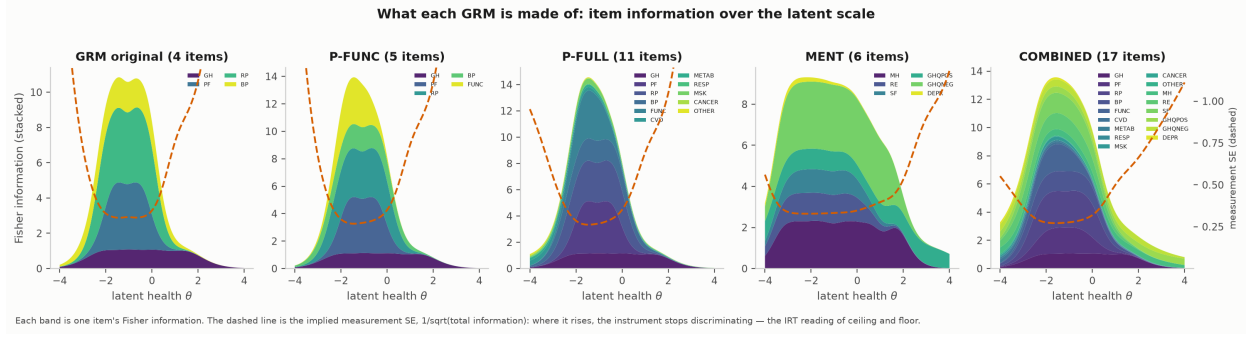


Figure 1: What each GRM is made of: stacked item information over the latent scale, with the implied measurement SE (dashed). FUNC adds a broad band of precision on the ill side; the condition items are visibly thin; GHQ-negative dominates the mental bank; the combined bank is tallest and widest but its SE still rises sharply above $\theta \approx +1$.

	P-FUNC	P-FULL	P-REC	MENT	COMBINED
max positive Q_3	-0.048	+0.144	+0.080	+0.153	+0.582 [†]
EAP identity ($\text{var} + E[\text{psd}^2]$)	1.000	1.000	1.000	1.000	0.999
r with original GRM θ	0.994	0.985	—	—	0.917

[†]the mental-pair clustering that is itself the two-dimensionality evidence; the physical side stays clean (§6).

Discriminations: the functioning items dominate — FUNC $a = 3.3$ (P-FUNC: GH 2.0, PF 3.7, RP 3.4, BP 2.4) — while the condition-group items are weak (CVD 1.15, MSK 0.85, RESP 0.68, METAB 0.59, CANCER 0.58, OTHER 0.62): once functioning is measured well, a diagnosis adds little about *current* physical health. On the mental side, the GHQ negative-wording testlet ($a = 3.15$) and MH (2.75) carry the bank; the ever-depression diagnosis is again weak (0.97).

4 How the measures relate

Correlations over all person-years (descriptives/01):

	θ phys (FULL)	θ ment	θ combined
original GRM (4 testlets)	0.985	0.48	0.92
UKHLS PCS	0.904	0.23	0.76
physical-only composite	0.946	0.48	0.90
UKHLS MCS	0.27	0.882	0.58
GHQ-12 Likert	-0.38	-0.904	-0.64
SF-6D utility	0.71	0.799	0.88
chronic count	-0.58	-0.21	-0.51
θ phys $\times$ θ ment		0.457	

Three reads. The physical GRM is the object the project already validated, with functioning attached. The mental GRM is the MCS/GHQ construct without the orthogonality artifact. SF-6D correlates *more* with the mental theta (0.80) than the physical one (0.71) — and its closest cousin is the *combined* bank (0.88): the SF-6D was a mixed measure all along.

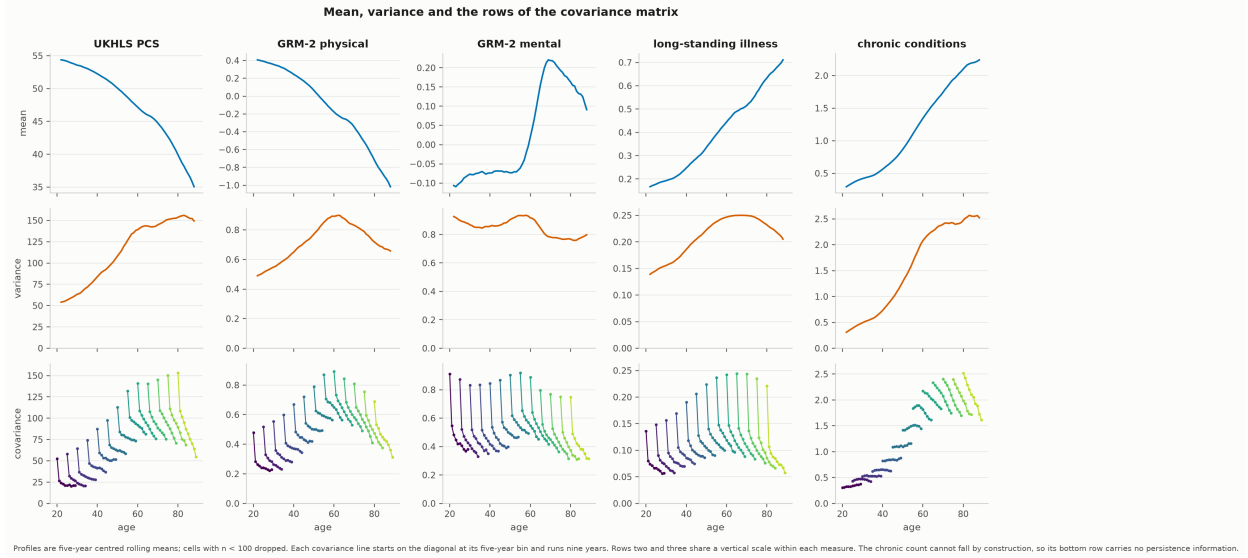


Figure 2: Mean, variance and the rows of the covariance matrix (the empirics note’s anatomy), rebuilt for the new measures. The mental GRM shows the textbook late-life wellbeing improvement; long-standing illness and the chronic count give the same physical anatomy from items disjoint from the SF-12.

Partial K-means (Figure 3): changing the *scoring* moves the typology modestly while changing the *construct* redraws it. The physical-only composite agrees with the baseline at ARI 0.67 (88% same cluster); the IRT re-spacings sit at ARI 0.43 (original GRM) and 0.41 (GRM-2 physical); SF-6D at 0.19 — and the mental GRM partitions people near-independently of the physical baseline (ARI 0.07; 49% same-cluster against a 47% chance rate). The physical typology is robust to how physical health is scored; the mental dimension is genuinely new information.

5 The ever-diagnosis question, resolved

The misgiving: historical diagnoses accumulate mechanically with age, so a diagnosis-bearing measure could manufacture “decline” out of stale paperwork. Four reads.

(1) The aggregate effect is real but bounded. $\mu(80) - \mu(30)$ is -1.117 SD functioning-only, -1.247 with ever-diagnoses, -1.170 with only diagnoses ≤ 10 years old. Diagnoses steepen the measured lifecycle decline by 0.130 SD (12%), and 59% of that extra steepness comes from diagnoses more than a decade old — leaving the stale-accumulation component at $\sim 6\%$ of the total decline.

(2) The potential was much larger than the realised effect. Half the ever-stock is >10 years old at every age past 40 (§2.4). The damage stays small because the estimator protects itself: items that cannot predict the functioning responses get low discriminations, so stale-heavy condition items simply carry little weight.

(3) The items cannot tell stale from fresh — which cuts both ways. The person-level penalty $\theta_{\text{full}} - \theta_{\text{func}}$ is flat in the age of the oldest diagnosis (-0.05 SD at 0–5 years, -0.075 at 30+). Read as measurement of *current* health, that is exactly the mechanism feared: no within-item discounting exists. It is an argument *against trusting ever-items as current-state measures*, and simultaneously the reason their damage is capped.

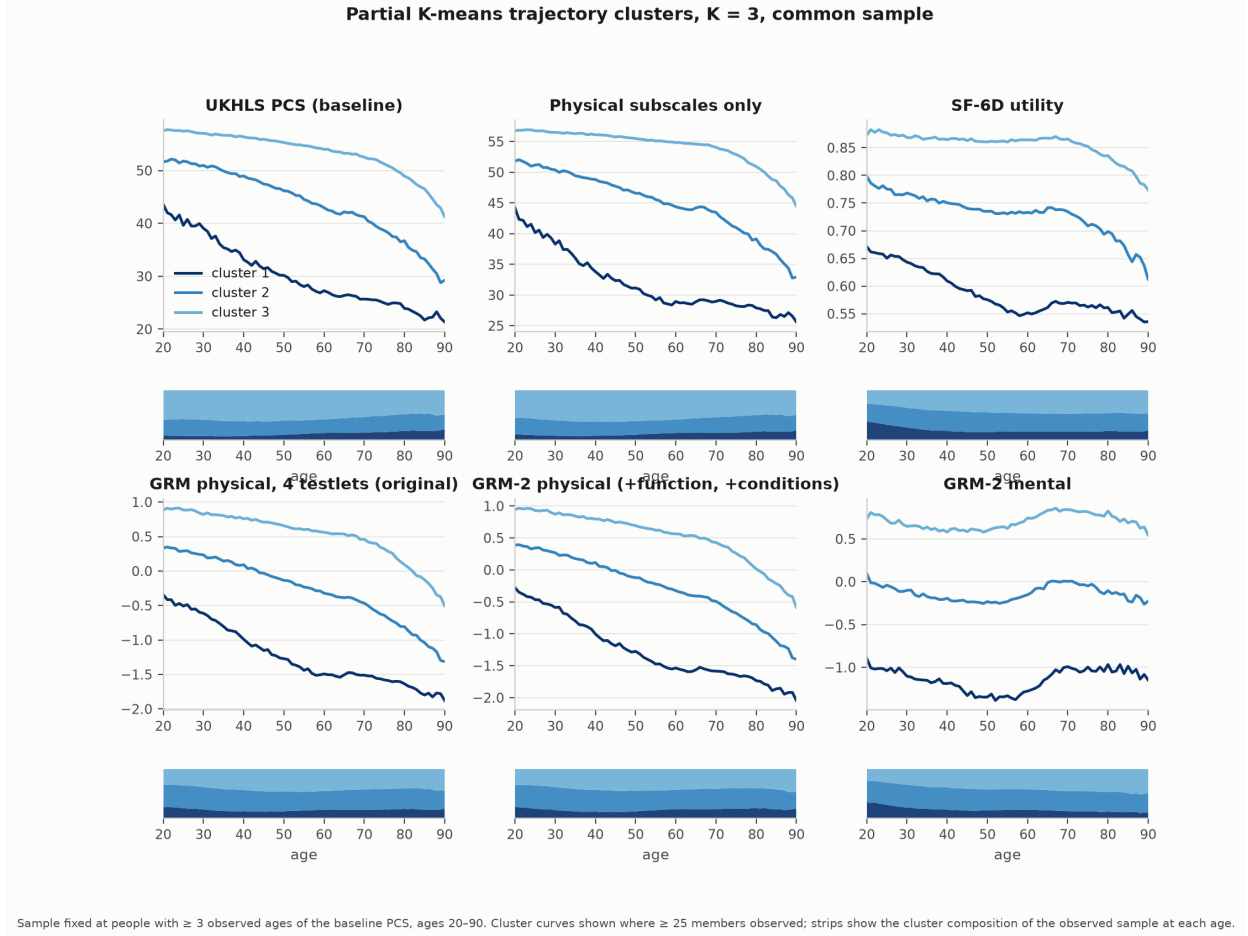


Figure 3: Partial K-means trajectory clusters ($K = 3$, 50 seeded inits, sample fixed at people with ≥ 3 observed baseline-PCS ages, ages 20–90).

(4) **Asked directly, the data mostly decline to discount.** P-TIMED splits every group into recent ($\leq 10y$) and stale ($> 10y$) items and lets the model estimate their separate discriminations — the in-framework analogue of an estimated recency-weight function:

Group	a recent	a stale	ratio
CVD	1.19	1.21	0.99
METAB	0.59	0.81	0.72
RESP	0.68	0.29	2.31
MSK	0.83	1.22	0.68
CANCER	0.59	0.56	1.05
OTHER	0.63	0.63	1.00

Only respiratory disease wants a discount (remitting childhood asthma). For arthritis and the metabolic conditions the *stale* item is more informative — progressive diseases, where an old diagnosis means longer duration and worse current health. The mechanistic-accumulation story is really a story about *remitting* conditions; for progressive conditions the old diagnosis is legitimately informative.

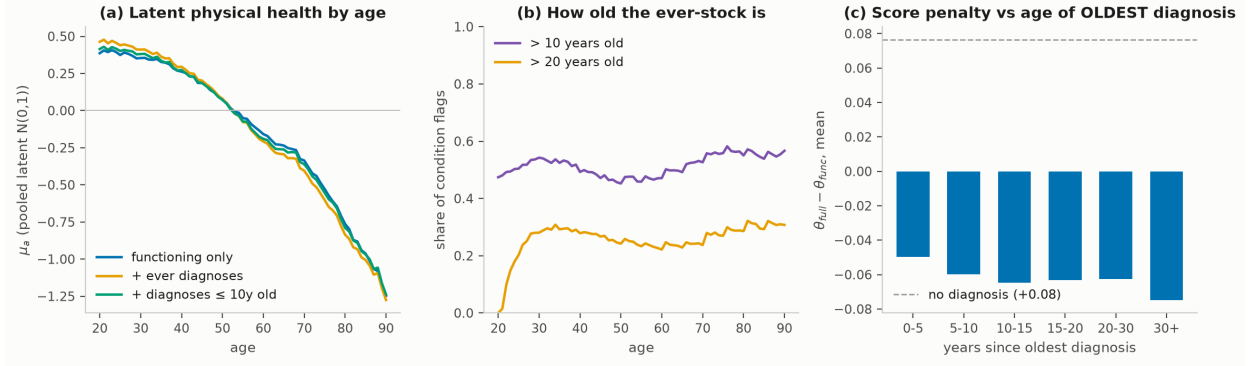


Figure 4: (a) latent age profiles with and without diagnoses and with only recent diagnoses; (b) the staleness of the ever-stock; (c) the score penalty against the age of the oldest diagnosis.

Related literature. Estimating recency weights on past exposures is the weighted-cumulative-exposure literature (Sylvestre & Abrahamowicz 2009; review in Kelly et al. 2024; R package WCE), with spline-estimated weight-by-recency functions — P-TIMED is its two-knot IRT analogue. Letting item parameters vary continuously with a covariate such as time since diagnosis is moderated nonlinear factor analysis (Bauer & Hussong 2009; regularised variants for DIF detection), the natural next refinement if a smooth weighting is ever needed. The simpler administrative-data practice is a fixed lookback window (Preen et al. 2006), which P-REC implements. Health-state valuation handles the same issue with acute vs. post-acute utilities for events like stroke.

Verdict. For lifecycle *dynamics*, use P-FUNC (or the original GRM, $r = 0.994$): a pure current-state instrument. P-FULL is a sound descriptive companion whose accumulation bias is now characterised and bounded (~ 0.08 – 0.13 SD over the lifecycle, concentrated in remitting conditions); P-REC and P-TIMED are its sensitivity band. Where conditions matter substantively, model them as separate channels in the Bayesian mixture, where “ever” is the correct state variable.

6 One dimension or two? The combined GRM

The combined bank (all seventeen items, one latent dimension) fits, scores, and correlates sensibly ($r = 0.92$ with the physical bank, 0.75 with the mental, 0.88 with SF-6D). But it fails the assumption the others satisfy: after conditioning on its single dimension, the mental items remain strongly positively correlated — $Q_3(\text{MH}, \text{GHQ}^-) = +0.58$, $Q_3(\text{GHQ}^+, \text{GHQ}^-) = +0.54$, $Q_3(\text{MH}, \text{RE}) = +0.33$ — while the physical side stays clean (max phys-phys $Q_3 +0.20$). That residual clustering is the data saying the mental items share a dimension the combined score cannot represent: with $r(\theta_P, \theta_M) = 0.457$, one dimension is a genuine misspecification, not a summary. The same criteria table shows the cost in substance: the combined score reintroduces a mental-health artifact of -0.82 SD at a fixed physical profile (the physical banks: 0.00), and its cohort spread collapses to 0.06 SD not because it is clean but because physical health improving and mental health deteriorating across cohorts *cancel* inside it — precisely the mixing the EIT note warned about. Use it only where a single index is externally imposed; a bifactor model (general + physical/mental group factors) would be the defensible one-number construction if that need becomes real.

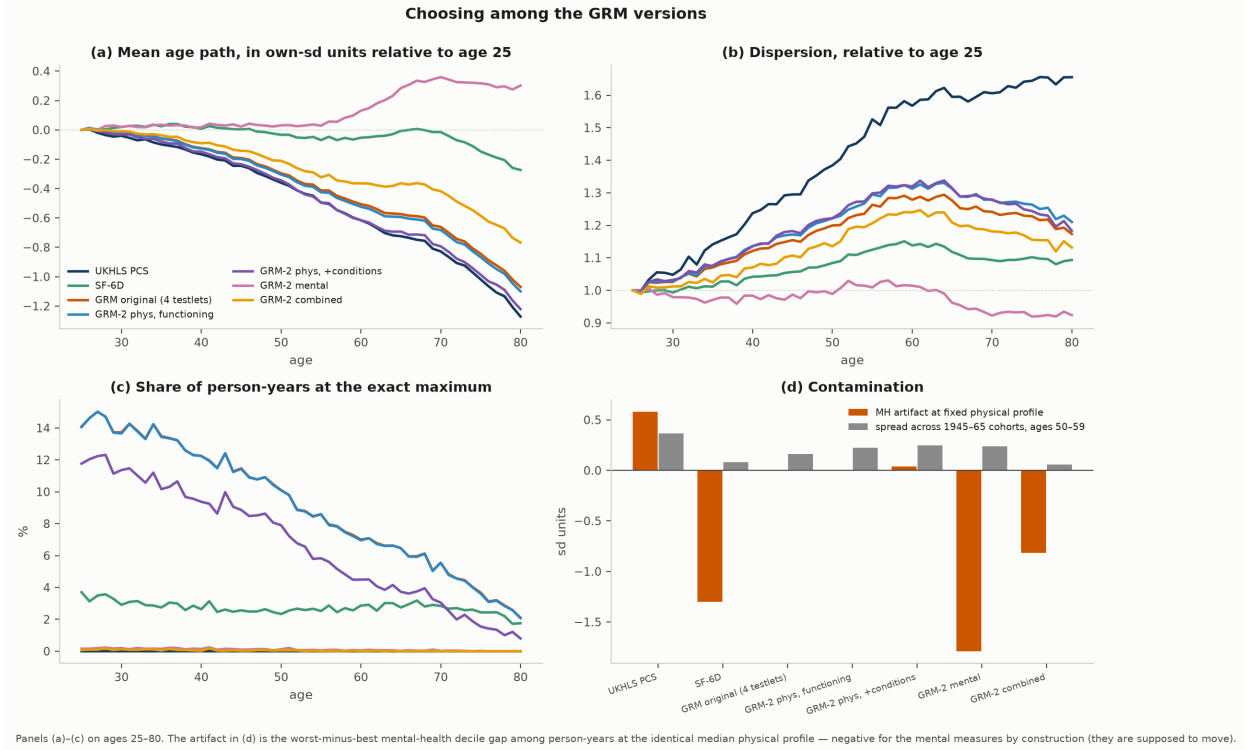


Figure 5: The measure-choice panel across GRM versions (the empirics note’s Figure 26, rebuilt): mean age path, fanning, top-censoring, and the two contamination reads.

7 Ceiling and floor

Two different questions hide under “ceiling effects”: how much probability mass sits at the best attainable response profile, and whether the instrument can *discriminate* among the healthy. More items help enormously with the first and only marginally with the second.

Measure	ceiling %	floor %	range (sd units)	psd at top	psd at bottom
UKHLS PCS (12 items)	0.89 ^a	~0	6.5	—	—
Physical-only composite	9.95	0.91	4.1	—	—
SF-6D utility	2.98	0.18	4.7	—	—
GHQ-12 Likert	0.20	0.31	6.4	—	—
GRM original (4 testlets)	9.55	0.97	4.4	0.65	0.45
P-FUNC	9.53	0.76	4.5	0.65	0.44
P-FULL	7.16	0.00	5.3	0.64	0.56
MENT	0.11	0.03	6.5	0.60	0.45
COMBINED	0.05	0.00	7.5	0.66	0.55

^ashare at the all-best 12-item profile (the exact-maximum read is degenerate for a continuous derived score). “psd” is the EAP posterior sd at the all-best / all-worst response pattern.

The optimistic half of the hypothesis is confirmed: ceiling mass collapses from 9.5% (original GRM) to 7.2% with conditions and to 0.05–0.11% for the mental and combined banks, and the floors of P-FULL and COMBINED empty entirely. Panel (c) of Figure 5 shows where it binds: the young end, where the physical banks still park 12–15% of person-years at the maximum.

The pessimistic half is also right, and Figure 1 shows why: the posterior sd at the very top is $\sim 0.6\text{--}0.66$ *for every version* — seventeen items buy no more precision among the healthy than four, because every added item has strongly negative thresholds (all these instruments measure ill-health). Extra items *spread* the top mass into a finer ranking (ties broken by GHQ and diagnoses) without adding information where the healthy live. Genuinely fixing the ceiling requires items with positive thresholds — vigorous-functioning items of the PROMIS physical-function type — which UKHLS does not carry. For the mixture models this is a manageable, known feature: the likelihood sees ordered categories, not the score, and the ceiling category is one of them.

The floor, in depth

The floor is the mirror image, and here the added items genuinely work — their negative thresholds sit exactly where the severely ill live (`descriptives/06, floor_depth.csv`):

	measurement SE at $\theta =$			min	% < -3	% at worst	distinct θ in
	-4	-3	-2	θ		pattern	bottom 1%
GRM original	2.15	0.73	0.33	-2.61	0.00	0.97	2
P-FUNC	2.12	0.67	0.30	-2.68	0.00	0.76	3
P-FULL	0.95	0.55	0.29	-3.38	0.03	0.00	575
MENT	0.56	0.34	0.33	-3.72	0.30	0.03	1,524
COMBINED	0.55	0.37	0.28	-4.43	0.24	0.00	3,708

The original GRM (and P-FUNC) have a real floor problem: the instrument goes dark below $\theta \approx -2.5$ (SE 0.7 at -3, 2.1 at -4), the scale simply ends at -2.6, and the entire bottom percentile collapses onto two or three response patterns — the worst-off are a spike, not a distribution. The condition items repair precisely this: their thresholds (-2.6 to -4.6) are the only ones deep in the ill tail, so P-FULL keeps SE at 0.55 at $\theta = -3$, extends the scale to -3.4, empties the worst-pattern spike entirely, and resolves the bottom percentile into 575 distinct positions. Among the P-FUNC bottom *decile* (37.6k person-years — the population a prevention analysis targets), P-FUNC distinguishes 1,252 values, P-FULL 16,522, COMBINED essentially one per person-year. The mental bank has the best floor of all (GHQ thresholds reach -3.5; SE 0.34 at -3).

This sharpens the ever-diagnosis trade-off of §5: the diagnosis items’ cost is a bounded accumulation bias in the age gradient; their benefit is measurement depth among the severely ill, which functioning items alone cannot provide because severe limitation saturates them. For analyses that turn on the worst-health group — describing the bottom class, targeting — P-FULL is the better physical instrument; for population dynamics, P-FUNC remains the clean choice.

8 Recommendations

1. Physical health for lifecycle dynamics: **P-FUNC** (or the original GRM). Mental health: **MENT**. Report both; their correlation (0.457) is a finding, not an assumption.
2. **P-FULL** as the descriptive companion when condition burden should show in the score; P-REC/P-TIMED quantify its recency sensitivity.
3. Conditions in the Bayesian mixture as separate channels; “ever” is the right state variable there.

4. Avoid the combined score unless a single index is externally required; if it is, build the bifactor version first.
5. SF-6D: treat as a mixed physical–mental utility (its closest GRM cousin is the combined bank), never as a physical measure; licence before publishing utilities.

9 Reproducibility

<code>data_cleaning/03_build_measures.py</code>	SF-12/SF-6D layer; validates against the construction note (max err 0.0054, 528,485 rows, the 6.6-pt artifact, tariff checks)
<code>data_cleaning/04_build_grm.py</code>	original GRM replicated to 10^{-5} on all 498,424 person-waves
<code>data_cleaning/05_build_chronic.py</code>	chronic layer; 100% timing coverage; $\geq$ EIT count on 100.000% of rows
<code>data_cleaning/06_build_grm2.py</code>	all GRM-2 specs; Q3, EAP identity, invariance, recency weights
<code>descriptives/01--06</code>	measure panel, correlations, the five figures, ceiling/floor table

Everything under `data/` derives from licensed UKHLS microdata and is never committed; all figures are aggregate statistics.

References

Barnett et al. (2012) *Lancet* 380:37–43. Bauer & Hussong (2009) *Psychol Methods* 14:101–25. Cella et al. (2010) *J Clin Epidemiol* 63:1179–94. Farivar, Cunningham & Hays (2007) *Health Qual Life Outcomes* 5:54. Goldberg et al. (1997) *Psychol Med* 27:191–7. Graetz (1991) *Soc Psychiatry Psychiatr Epidemiol* 26:132–8. Hankins (2008) *Clin Pract Epidemiol Ment Health* 4:10. Johnston, Propper & Shields (2009) *J Health Econ* 28:540–52. Kelly et al. (2024) *Pharmacoepidemiol Drug Saf* 33:e5701. Kriegsman et al. (1996) *J Clin Epidemiol* 49:1407–17. Mitnitski, Mogilner & Rockwood (2001) *ScientificWorldJournal* 1:323–36. Preen et al. (2006) *J Clin Epidemiol* 59:940–6. Samejima (1969) *Psychometrika Monogr* 17. Searle et al. (2008) *BMC Geriatr* 8:24. Sylvestre & Abrahamowicz (2009) *Stat Med* 28:3437–53. Wainer & Kiely (1987) *J Educ Meas* 24:185–201. Ware, Kosinski & Keller (1996) *Med Care* 34:220–33. Yen (1993) *J Educ Meas* 30:187–213.